

Annotated Bibliography

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[4] The article provides a summary of techniques presently used in sustainable development within the scientific community. The study begins by discussing the importance of sustainable development, the frequency of various techniques used to analyze sustainable development, and important issues in sustainable transportation, such as logistics and environmental issues. It then dives into the techniques that are currently being used to analyze sustainable development, which includes simulation, optimization, machine learning, and fuzzy set analysis. The Optimization section covered the history of optimization used in transportation and the variety of applications for optimization, including in the difference between urban transportation systems and in global/regional transportation systems. Applications in simulation focus on the growing integration of simulation and optimization models and on usages in logistics, traffic congestion, and on the usage of demand-based transportation systems (ride sharing applications). The machine learning section primarily summarizes a paper that uses a random forest model to understand patterns in a bike sharing transportation system, as well as touches on a variety of techniques (random forest, logistic regression, neural networks, support vector machines, decision trees, heuristics, multi-agent systems, and multi-variate analysis) for a variety of analytical, traffic-related outcomes. The section on fuzzy sets takes note of the wide range of uses that recent projects have undertaken utilizing fuzzy sets. The paper notes that sustainability factors into many of the research projects, where businesses seek sustainability to increase innovation and enhance competitiveness. The paper concludes by suggesting that more of the investigated technologies be applied in tandem with one another to increase effectiveness. The section on machine learning will be the most relevant to the IS, but the other sections provide relevant background information about other advances in the field as well as ways that these tools are used in the business world.

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[2] This chapter provides a brief overview of the various techniques that combine GIS data and machine learning currently employed in the civil engineering field. While it lists the name of the technique and a very brief paragraph about its recent advances, its upsides, and its downsides, the paper does not explain what the technique is or when it would be most beneficial to use it. There were also typos and incomplete sentences present in the writing. Neural networks, random forest and gradient boosting, support vector machines, decision trees, bayesian networks, gaussian processes, and ensemble learning all receive paragraph-long summaries. The paper then describes applications of machine learning to GIS, primarily in the analysis of environmental hazards (flood prediction, groundwater modeling,

erosion, landslip susceptibility assessment, and LULCC) but also in areas such as public health, maintaining biodiversity, or in analyzing infrastructure costs. The paper concludes with an assessment that the combined use of GIS data and machine learning methods have the potential to benefit a large variety of fields. Although the paper encompasses a breadth of information about machine learning and GIS, it is held back by its brevity and will be more useful to me as a list of keywords to research in-depth elsewhere.

References

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